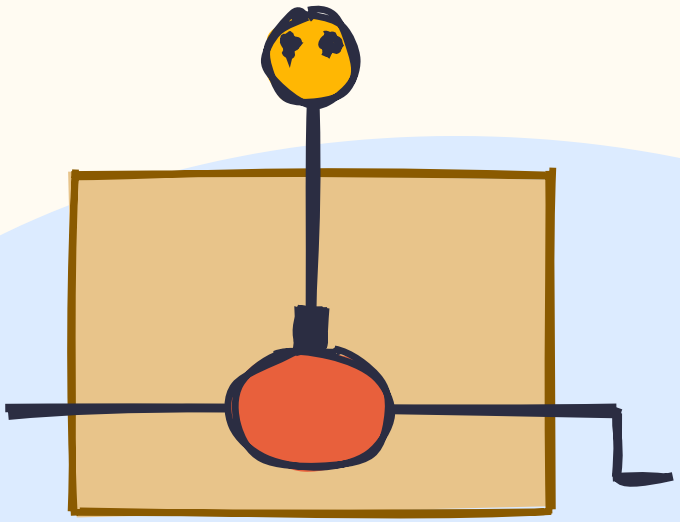
Little Explorers · Class 28

Cardboard Automaton

Turn the handle and watch it come alive!

Cranks | Cams | Moving toy



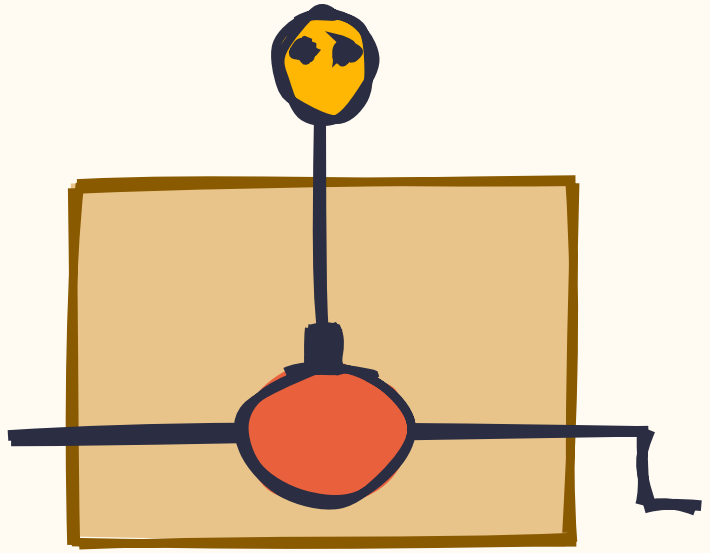
Did you know?
Automatons are moving toys that people have built for hundreds of years, long before computers!

Let's wonder...

Can a box make a toy move with no batteries?

Magic box?

Turn the handle and the little bird on top flaps up and down! There are no batteries and no computer. What's hiding inside the box?



Batteries?

A tiny robot?

Clever shapes!

Today: you build a **cardboard automaton**: a toy that moves using a crank and a cam.

LEARN

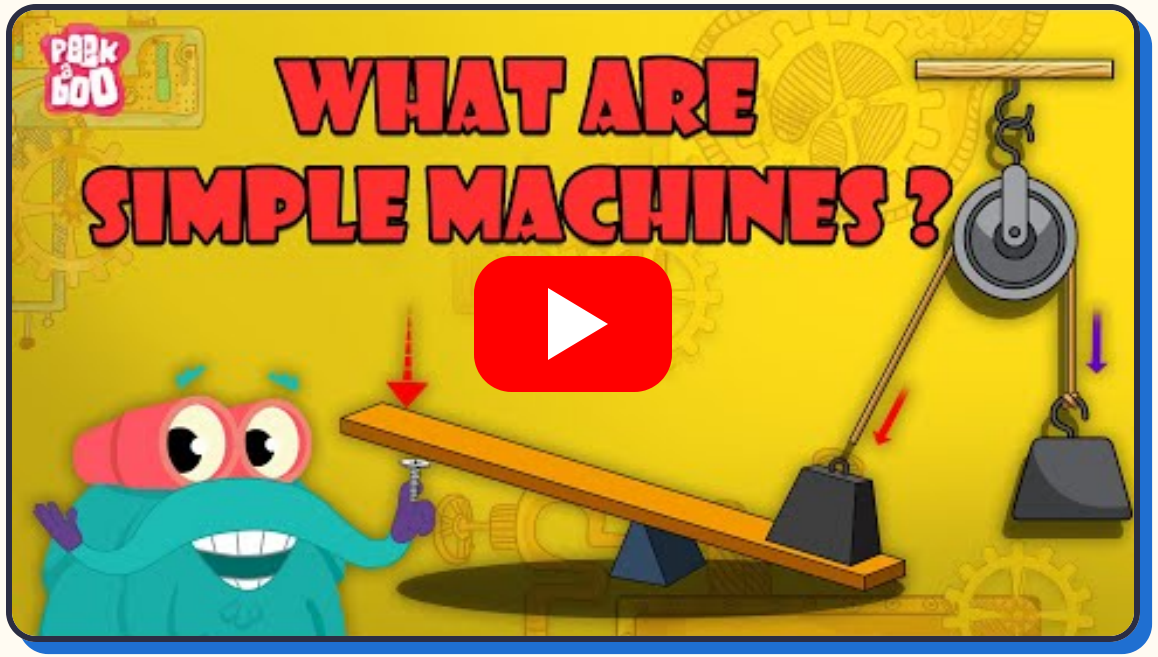
Learning objectives

Today I can…

1		**Crank turns:** say that a **crank** turns round and round	**Proof:** I turned the crank and saw it spin
2		**Cam lifts:** show that a **cam** (bumpy wheel) makes something go **up and down**	**Proof:** my cam made the stick go up and down
3		**Moving toy:** build a **cardboard toy** that moves when I turn the handle	**Proof:** my toy moves!

Objective 1 · Crank turns · video

Video time!



What Are Simple Machines?

The Dr. Binocs Show · about 6 minutes · youtu.be/tk9iUjMEnaY

Watch for...

1. What is a **simple machine**?
2. Name a **wheel and axle**.
3. How do machines make work easier?

Scan to watch

After the video: Turn your wrist round and round like a crank. What could that turn make move?

LEARN

Objective 1 · Crank turns

A crank turns round and round

The idea

A **crank** is a handle on a stick. Turn the handle and the stick (the **axle**) spins round and round.

It's a **wheel and axle**: a simple machine!

Example: a pencil sharpener



A handle sharpener: turn the handle, and the blade inside spins round. That's a crank!

Objective 2 · Cam lifts

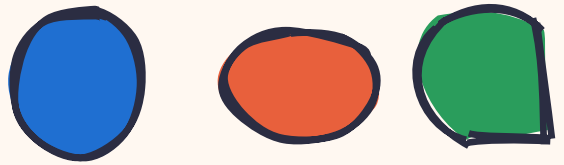
A cam turns round into up and down

The idea

A **cam** is a wheel that is **not quite round**. When it spins, the bumpy part **pushes up** a stick resting on it, then lets it fall.

Round and round becomes **up and down**!

Example: a bumpy road



round egg snail

Ride a bike over a bump and you go UP, then DOWN. A cam is a bump that comes round again and again!

Real world

Cams and cranks at work



Car engines
Cams open and close tiny doors in the engine



Music boxes
Turn the crank, the song plays



Sewing machines
Cams push the needle up and down



Toys
Wind-up toys hop and flap

? What toy at home moves up and down when you turn or push something?

Old inventions
Long ago, clever inventors built moving birds and dolls with just cams and cranks!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human cam
2. Build the box
3. Add the cam and crank
4. Add your character
5. Cam experiment
6. Inventor challenges

Class game

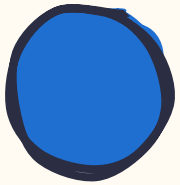
Human cam

How to play

1. One child is the **cam**: walk slowly in a circle holding an arm out.
2. Another child is the **stick**: stand inside the circle with a hand on the cam's arm.
3. When the arm goes up, the stick stands tall; when it goes down, crouch!
4. Speed up!

Then discuss

Did the stick go up and down while the cam went round?



Big idea

Round-and-round can make up-and-down.

MAKE

Objective 1 · Build step 1 of 3

Build the box

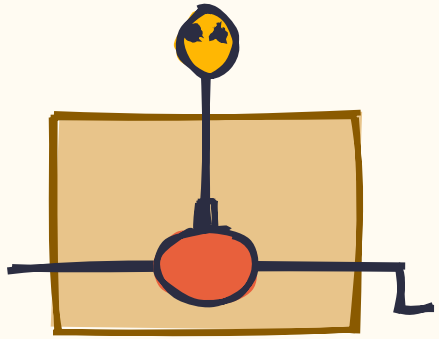
a Take a small cardboard box (a tissue box is perfect).

b Teacher makes two holes on opposite sides, the same height.

c Push a **skewer** through both holes: that's the axle.

d Make one hole in the **top** of the box, right above the middle of the axle.

e Check the skewer spins freely.



Check: The axle spins when you turn it.

Skewers are sharp! The teacher makes the holes.

MAKE

Objective 2 · Build step 2 of 3

Add the cam and crank

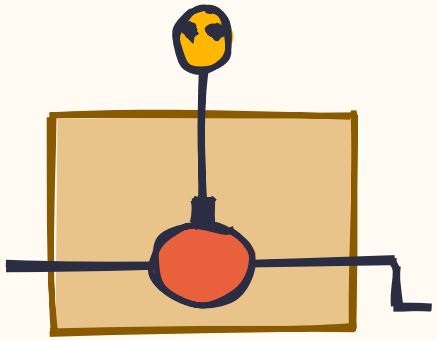
a Cut an **egg-shaped** cam from thick card. Make a hole off-centre.

b Push the cam onto the axle, inside the box, under the top hole.

c Bend a straw or card strip into a **handle** on the outside end.

d Put a straw through the top hole so it rests on the cam.

e Turn the handle: does the straw go up and down?



Check: The straw goes up and down when you turn the handle.

Objective 3 · Build step 3 of 3

Add your character

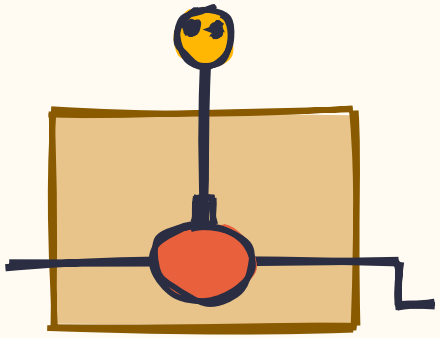
a Draw a character on card: a bird, a fish, a rocket, a jumping frog!

b Colour it and cut it out.

c Glue it to the top of the straw.

d Decorate the box like a scene.

e Turn the handle and tell a story!



Check: Your character moves up and down.

Experiment!

Cam experiment

Cam shape	I think it will…	What happened
Round cam (hole in middle)	move / stay still?	_____
Egg cam	small / big move?	_____
Snail cam	smooth / sudden drop?	_____

Talk about it

Which cam made the biggest bounce? Which was the funniest?

Swap cams

Make two cams and swap them to compare!

Level up!

Inventor challenges

Choose your challenge



Two cams: add a second cam and a second character.



Spinner: make a character that spins round instead.



Story show: turn your automaton and tell its story.



Motor: could a motor turn the crank for you?

Think like an engineer

Why does a round cam with a hole in the middle not move the stick?

Boss level

Make a character that flaps its wings!

Objective check

Can I really do it? Prove it.

01	Show me how a crank turns.	○ ○ ○
02	Show how your cam makes the stick go up and down.	○ ○ ○
03	Turn your automaton. What does it do?	○ ○ ○

Rate yourself
Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Automaton museum: display every machine with a name card.

Show what you know

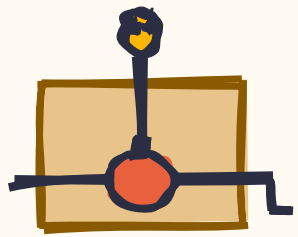
What did we learn?

- [x] A crank turns an axle round and round
- [x] A cam is a not-quite-round wheel
- [x] Cams turn round-and-round into up-and-down
- [x] Simple machines can make toys move

Exit ticket: what does a cam do?

At home

Find a wheel and axle at home: a door handle, a tap, a bicycle!



Next class: Teach the Computer Our Sounds!